

HOMESTEAD ACADEMIC MAPPING & UNIT STUDIES GUIDE

K-12 Multi-Level Curricular Extension • Missouri Compliance Series

Curricular Integration Overview

Homeschooling with Necessity operates on a foundational, real-world instruction model: **Real-Life Work yields Multi-Subject Academic Outcomes**. Under Missouri Revised Statute **RSMo § 167.012**, home educators must maintain chronological logs documenting 1,000 hours of annual instruction, with at least 600 hours allocated to core subject areas (Mathematics, Science, Language Arts, Social Studies) and 400 of those core hours occurring directly at the primary home location. By utilizing this multi-level unit plans guide, parents can translate daily agrarian duties and physical work into verified academic portfolios across all age levels—from early childhood through high school.

Academic Unit	Main Core Discipline	Primary High School / MS Core	Grades 2-4 Adaptation	Grades K-2 Adaptation
Unit 1: Pantry Audit	Mathematics & Economics	Unit rates, algebra, FCR, waste percentage formulas.	Addition, basic fractions, category graphing, smart snacks.	One-to-one sorting, counting, shape/color matching, verbal sharing.
Unit 2: Water Catchment	Physical Science & Lab	Fluid dynamics, bio-filters, microbiology assays, pH chem.	Weather diary, sand/gravel property sort, settlement logs.	Wet vs. dry play, sink/float, mechanical flow, outdoor cycle walk.

K-12 Portfolio Requirement Notice: Regardless of grade level, Missouri law mandates the retention of representative portfolios containing dated chronological logs, academic coursework sheets, and physical samples of completed work. Use this guide alongside your **Missouri Hours Log Spreadsheet** to preserve audit-ready compliance folders for every student.

UNIT 1: THE HOMESTEAD PANTRY AUDIT

Core Subject Focus: Applied Mathematics, Statistics & Family Economics

A. Core Level (Grades 5-8 & High School Path)

Objective: Students master operational statistics, ratios, and algebraic calculations to manage actual family food assets. This level demands precise mathematical tracking and calculation of unit rates to calculate exact yearly savings of homestead preserves.

- **Step 1: Ingest & Inventory:** Audit the home pantry, creating structured logs of item counts, food types (grains, cans, preserves), packaging weights, and expiration dates.
- **Step 2: Unit Cost Calculations:** Calculate price-per-ounce or price-per-gram for 5 main bulk products by dividing total cost by package volume.
- **Step 3: Proportion Analysis & Data Plots:** Map inventory proportions onto a hand-drawn circle chart or digital pie graph, demonstrating ratios.
- **Step 4: Cost Savings Projections:** Determine a frequently wasted grocery item (e.g., canned beans or pickles) and calculate the yearly monetary difference of replacing it with home-grown, pressure-canned homestead alternatives.

B. Adapted Level (Grades 2-4 Path)

Objective: Introduce basic addition/subtraction, counting units, physical measurements, simple fractions, and pictographs. The mathematical complexity shifts from advanced unit rate divisions to foundational group comparisons.

- **Step 1: The Inventory Hunt:** Count home pantry items up to 50. Categorize by simple containers: cans, cardboard boxes, and jars.
- **Step 2: Ounces & Weights:** Read labels to locate the 'Net Weight' (ounces or grams) on three food items. Put them in order from lightest to heaviest.
- **Step 3: Block Graphing:** Draw a block graph on grid paper where each square represents 1 item, comparing visual quantities across categories.
- **Step 4: Smart Snacking:** Count snack items that are wasted in a typical week. Subtract that number from total snacks bought to see how many were actually eaten.

C. Adapted Level (Grades K-2 Path)

Objective: Establish counting to 20, visual sorting, identifying physical attributes (color, size, shape), and verbal storytelling. Focus on hands-on matching and developing a concrete connection between food on the farm and food in the home.

- **Step 1: Shape & Color Sort:** Group pantry packages by color (e.g., green cans, yellow boxes) or basic 3D geometry (cylinders vs. rectangular boxes).
- **Step 2: One-to-One Correspondence:** Count aloud each item in a pile up to 10 or 20, pointing to each package to build tracking skills.
- **Step 3: Picture Graph:** Draw circles on paper matching the counts (e.g., 4 green circles for cans, 6 orange squares for boxes).

- **Step 4: The No-Waste Kitchen Story:** Verbally discuss how fruits and vegetables grow on a farm. Draw a colorful picture of their favorite pantry food.

UNIT 2: RAINWATER CATCHMENT & FILTRATION TEST

Core Subject Focus: Earth Science, Physics, Chemical Properties & Lab Assays

A. Core Level (High School & Adult Path)

Objective: Apply physics and chemistry to build and analyze an off-grid mechanical and bio-filtration system. Students observe the clearance of turbid sediment water using cost-effective homestead components (recycled buckets, gravel, sand, and activated charcoal). Requires written lab reports and microscopic or biological evaluations of filtration success.

- **Step 1: Physical Design:** Calculate fluid volume, gravel-to-sand height ratios, and lay out layers to optimize gravity-fed filtration.
- **Step 2: Budgeting Restrictions:** Assemble components keeping total construction costs under \$20 by using recycled buckets and natural materials.
- **Step 3: Sediment & Clarity Test:** Collect turbid runoff and measure suspended sediment. Run water through the layers and record results.
- **Step 4: Non-Potable Applications:** Run pH test assays to determine water safety bounds for garden irrigation and non-drinking livestock usage.

B. Adapted Level (Grades 2-4 Path)

Objective: Introduce weather tracking, investigating physical properties of earth materials, and simple visual logging. Students study the separation of solids and liquids and the roles of natural elements in water cycle filtration.

- **Step 1: Rain Gauge Diary:** Place a clear jar outside during rain and use a ruler to measure and write down the daily precipitation heights.
- **Step 2: Earth Material Sort:** Touch and classify filter ingredients. Sort particles into three jars: big pebbles (gravel), tiny pieces (sand), and black flakes (charcoal).
- **Step 3: Mini-Filter Science:** Observe water passing through a clear plastic bottle filter layer. Draw a picture of the muddy pre-filter water next to the clearer post-filter water.
- **Step 4: The Plant Drink:** Help pour filtered rainwater onto vegetables or flowers, recording how moisture acts as fuel for plant growth.

C. Adapted Level (Grades K-2 Path)

Objective: Foster sensory-based science, fine motor pouring skills, cause-and-effect observations, and basic nature appreciation. Focus on early tactile discovery and exploration of physical states.

- **Step 1: Wet vs. Dry Sensory Play:** Touch dry sand and dry pebbles. Pour water on them and talk about the sensory changes (dry/wet, heavy/light, dusty/muddy).
- **Step 2: Sink or Float Mystery:** Drop a pebble, a piece of charcoal, and a leaf into water. Watch and identify which objects sink to the bottom and which float.
- **Step 3: Visual Sifting Fun:** Pour water mixed with large leaves and twigs over a screen or kitchen colander. Watch how the mesh catches big debris while water falls through.
- **Step 4: Rainwalk Discovery:** Walk outside after a rainfall. Find where mud forms, where water pools, and search for earthworms or damp soil in the garden.

AUDIT-PROOF PORTFOLIO TRACKING MATRIX

K-12 Evidence-Based Standards by Developmental Band

Grade Band	Required Log Type	Portfolio Artifact Samples	Core Hours Focus Areas
Grades K-2	Simple verbal overview checkmark log checked by parent.	Drawings of sorted shapes, photo of sensory water play, short dictated sentences.	Visual counting, tactile properties of matter, oral expression, spatial reasoning.
Grades 2-4	Parent-guided digital/manual hours log with category checkboxes.	Hand-drawn pantry bar graphs, rain gauge measurements, physical material sort logs.	Addition/subtraction word problems, weather patterns, states of matter, basic sorting.
Grades 5-8	Student-led manual logging sheet reviewed weekly by parent.	Detailed bulk food spreadsheets, proportion math sheets, filtration design drafts.	Ratios & decimal math, biological decomposing variables, local agrarian history.
High School	Formal Missouri Hours Log Spreadsheet with automated formulas.	Unit cost algebra sheets, detailed rainwater catchment lab reports with pH assay logs.	Fluid dynamics equations, microbiology chemistry, double-entry farm accounting.

Continuous Development & Community Feedback

This curriculum guide represents an ongoing, dynamic resource built specifically for families aiming to balance practical agrarian homesteading with clear, structured Missouri state compliance. We are actively seeking parental feedback to refine these lesson studies, build out more units, and provide further downloadable tools like our digital Excel spreadsheets. Please send your feedback, lesson observations, and student success stories directly to our support desk as we work together to build the ultimate self-directed K-12 learning ecosystem.